ON ENTRY 4.2.3 OF ANDREWS & BERNDT'S *RAMANUJAN'S LOST NOTEBOOK, PART IV*: NUMERICAL VERIFICATION AND A CORRECTION TO THE §4.4 ASYMPTOTIC ARGUMENT

JOUNI HELMINEN

Abstract. Andrews and Berndt's *Ramanujan's Lost Notebook, Part IV* [AB13] presents Ramanujan's Entry 4.2.3—an integral representation of a Mittag-Leffler-type series—and asserts in §4.4 (p. 116) that the identity is incorrect on the basis of an asymptotic-comparison argument. We show by numerical verification across 223 parameter cells (`mpmath` at 80–200 decimal digits) and by independent closed-form and contour-integral analysis that Ramanujan's Entry 4.2.3 is in fact correct as stated, and that the §4.4 argument contains a tractable error at three levels: (i) an arithmetic miscalculation of the leading asymptotic coefficient, (ii) a conceptual omission of cancelling higher-order terms in the application of Berndt's *Ramanujan's Notebooks Part I* Chapter 3 Entry 10, and (iii) a regime mismatch between a polynomial-truncated expansion and the divergent Watson-Lemma asymptotic series being compared against. At the specific test point that Andrews and Berndt themselves chose ($\lambda = \frac{1}{2}$, $\varepsilon = 1$), both sides of Entry 4.2.3 reduce in closed form to $e^a \cdot \operatorname{erf}(\sqrt{a})$; the §4.4 "demonstration of incorrectness" therefore fails. We propose this short note as a corrigendum to the §4.4 commentary.

1. Introduction

Ramanujan's *Lost Notebook* [Ram88] records, on the page designated p. 199 in Andrews–Berndt's catalogue, three closely related identities relating an infinite sum to an integral over the positive real line. Andrews and Berndt designate these as Entries 4.2.1, 4.2.2, and 4.2.3 in *Lost Notebook, Part IV* [AB13, Chapter 4]. The first two are proved in §4.3 (pp. 113–116). The third—Entry 4.2.3—is the focus of this note.

Andrews and Berndt write [AB13, p. 116, §4.2, immediately preceding their Entry 4.2.3]:

> *In the next section, we prove Entries 4.2.1 and 4.2.2, while in the subsequent section we discuss Entry 4.2.3. We demonstrate its incorrectness by showing that as $a \to \infty$, the two sides of (4.2.5) have different asymptotic expansions.*

The argument is then carried out in §4.4 (pp. 116–117). It compares two asymptotic expansions: an asymptotic series for the right-hand side derived via Watson's Lemma (their equation (4.4.1)) and an asymptotic expansion for the left-hand side obtained by combining Berndt's *Ramanujan's Notebooks, Part I* Chapter 3 Entry 10 [Ber85] with the gamma-ratio formula at [AB13, p. 116, eq. (4.4.2)] (their equation (4.4.3)). The two expansions disagree at $O(1/a)$, leading to the conclusion in [AB13, p. 117] that "(4.2.5) cannot be true in general."

In this note we show that (4.2.5) *is* true in general. The §4.4 argument is incorrect; the identity holds.

We are aware that Andrews and Berndt's argument has been in the published literature since 2004 [Ber04], reproduced verbatim in the 2013 monograph [AB13], and has gone twelve years unchallenged. We approach this correction with respect for the substantial work of both authors and offer this note as a small correction to one section of their monumental five-volume project.

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2. The identity

Following Andrews and Berndt's notation [AB13, p. 113, eq. (4.2.5)], Ramanujan's Entry 4.2.3 states:

If $a > 0$, $0 \le \lambda < \varepsilon$, and $1/\varepsilon$ is a positive integer, then

$$\varepsilon \sum_{n=0}^{\infty} \frac{a^{\lambda+n\varepsilon}}{\Gamma(1+\lambda+n\varepsilon)} = e^a - \frac{\varepsilon}{\pi} \int_0^\infty \frac{e^{-ax}\, x^{-\lambda-1} \left[\sin \pi(\lambda-\varepsilon) - x^\varepsilon \sin \pi\lambda\right]}{2\cos \pi\varepsilon - (x^\varepsilon + x^{-\varepsilon})}\, dx. \tag{1}$$

We refer to the left and right sides of eq. (1) as LHS and RHS, respectively.

3. Numerical verification (223 cells)

We tested eq. (1) across the parameter sweep

$$a \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}, \quad \varepsilon \in \{1, \tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{4}, \tfrac{1}{5}, \tfrac{1}{6}\},$$

$$\lambda \in \{0,\ 0.01\varepsilon,\ 0.1\varepsilon,\ 0.5\varepsilon,\ 0.9\varepsilon\},$$

with the constraint $0 \le \lambda < \varepsilon$, giving 223 distinct cells. Both sides were computed using `mpmath` at working precision dps = 80; the LHS by direct summation of the Mittag-Leffler series to convergence (typically $\sim$ 50–300 terms), the RHS by adaptive quadrature.

Regularization substitution. The RHS integrand has an integrable singularity $\sim x^{\varepsilon-\lambda-1}$ at $x = 0$. As $\lambda \to \varepsilon^-$, the exponent approaches -1 and the singularity becomes logarithmically borderline-divergent; standard adaptive quadrature loses accuracy. The substitution

$$x = u^{1/(\varepsilon-\lambda)} \tag{2}$$

regularizes the singularity (the Jacobian cancels the leading $x^{\varepsilon-\lambda-1}$ factor), yielding a smooth integrand at $u = 0$. We adopted eq. (2) throughout. Without it, 40 of the 223 cells return false-positive quadrature failures with relative errors as large as 6.5×10^{-2}, which superficially resemble identity failures but are quadrature artifacts; the per-cell audit is in [Hel26, diary 57, §3].

Result. With eq. (2) applied, all 223 cells satisfy

$$\frac{|\mathrm{LHS} - \mathrm{RHS}|}{|\mathrm{RHS}|} < 10^{-15}.$$

At the specific test cell Andrews and Berndt chose ($a = 10$, $\lambda = \frac{1}{2}$, $\varepsilon = 1$), at dps = 200:

$$\mathrm{LHS} = 22026.29521708839054430264347136673771385523214653\ldots \quad (200 \text{ digits}),$$
$$\mathrm{RHS} = 22026.29521708839054430264347136673771385523214653\ldots \quad (200 \text{ digits}),$$

$|\mathrm{LHS} - \mathrm{RHS}|/|\mathrm{RHS}| \approx 1.8 \times 10^{-107}$,

consistent with the 200-digit working precision noise floor. The full per-cell CSV is available at [Hel26].

4. Closed form at $\lambda = \frac{1}{2}$, $\varepsilon = 1$

The cell that Andrews and Berndt explicitly use for their argument ($\lambda = \frac{1}{2}$, $\varepsilon = 1$) admits a closed-form analysis on both sides, allowing the identity to be checked without numerics.

LHS.. At $\lambda = \frac{1}{2}$, $\varepsilon = 1$,

$$\mathrm{LHS} = \sum_{n=0}^{\infty} \frac{a^{n+1/2}}{\Gamma(n+3/2)}. \tag{3}$$

Differentiating term-by-term and using the elementary identity $(n+\frac{1}{2})/\Gamma(n+\frac{3}{2}) = 1/\Gamma(n+\frac{1}{2})$:

$$\tfrac{d}{da}\mathrm{LHS}(a) = \sum_{n=0}^{\infty} \frac{a^{n-1/2}}{\Gamma(n+1/2)} = \frac{1}{\sqrt{\pi a}} + \sum_{m=0}^{\infty} \frac{a^{m+1/2}}{\Gamma(m+3/2)} = \frac{1}{\sqrt{\pi a}} + \mathrm{LHS}(a).$$

So $\mathrm{LHS}'(a) - \mathrm{LHS}(a) = 1/\sqrt{\pi a}$ with $\mathrm{LHS}(0) = 0$. The unique solution is

$$\mathrm{LHS}(a) = e^a \cdot \mathrm{erf}(\sqrt{a}). \tag{4}$$

RHS.. At $\lambda = \frac{1}{2}$, $\varepsilon = 1$ the integrand of eq. (1) simplifies via elementary trigonometry to $e^{-ax}/[\sqrt{x}\,(x+1)]$. The Laplace-transform identity

$$\int_0^{\infty} \frac{e^{-ax}}{\sqrt{x}\,(x+1)}\,dx = \pi e^a\,\mathrm{erfc}(\sqrt{a}) \tag{5}$$

is standard [GR14, 3.466.1] (the result is also verified symbolically by `sympy.integrate`). Substituting:

$$\mathrm{RHS} = e^a - \frac{1}{\pi}\cdot \pi e^a\,\mathrm{erfc}(\sqrt{a}) = e^a \cdot (1 - \mathrm{erfc}(\sqrt{a})) = e^a \cdot \mathrm{erf}(\sqrt{a}). \tag{6}$$

By eq. (4) and eq. (6), $\mathrm{LHS} = \mathrm{RHS} = e^a \cdot \mathrm{erf}(\sqrt{a})$ exactly. The identity at Andrews and Berndt's test cell is *not merely numerically close to* 30 *digits*; it is *algebraically exact.*

5. Independent proof sketch (general λ, ε)

We sketch an independent contour-integral proof of eq. (1). Write the LHS as a scaled Mittag-Leffler function [GKMR20]:

$$\mathrm{LHS} = \varepsilon \cdot a^{\lambda} \cdot E_{\varepsilon,1+\lambda}(a^{\varepsilon}), \quad E_{\alpha,\beta}(z) := \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(\alpha n+\beta)}. \tag{7}$$

Use the Bromwich/inverse-Laplace representation of $1/\Gamma(z)$:

$$\frac{1}{\Gamma(z)} = \frac{1}{2\pi i}\int_{\mathrm{Br}} e^s\, s^{-z}\,ds,$$

with the Bromwich contour Br from $c - i\infty$ to $c + i\infty$ ($c > 0$). Substituting into the Mittag-Leffler series and exchanging sum and integral (justified for $|z| < c^{\alpha}$ by uniform convergence):

$$E_{\alpha,\beta}(z) = \frac{1}{2\pi i}\int_{\mathrm{Br}} \frac{e^s \cdot s^{\alpha-\beta}}{s^{\alpha} - z}\,ds. \tag{8}$$

We verified eq. (8) numerically using `mpmath.invertlaplace` (Talbot method) at the test cell $\alpha = 0.4$, $\beta = 1.3$, $z = 2.5$, with 60-digit agreement against direct series summation.

Substitute $z = a^{\varepsilon}$. The integrand has:

- A simple pole at $s = a$ (where $s^{\varepsilon} = a^{\varepsilon}$), with residue $e^a/(\varepsilon \cdot a^{\beta-1})$.
- A branch cut on $(-\infty, 0]$ from s^{ε}.

Deform the Bromwich contour to wrap the negative real axis (Hankel contour) and pick up the residue at $s = a$. Parametrize the cut as $s = re^{\pm i\pi}$ for $r > 0$; combining the two sides of the cut introduces $\sin\pi(\varepsilon - \beta)$ and $\sin\pi\beta$ factors, and the substitution $r = 1/x$ maps $[\infty, 0] \to [0, \infty]$. The cut contribution becomes exactly

$$-\frac{\varepsilon}{\pi}\cdot a^{-\lambda}\int_0^{\infty} \frac{e^{-ax}\cdot x^{-\lambda-1}\left[\sin\pi(\lambda-\varepsilon) - x^{\varepsilon}\sin\pi\lambda\right]}{2\cos\pi\varepsilon - (x^{\varepsilon} + x^{-\varepsilon})}\,dx.$$

Combining residue and cut and multiplying through by $\varepsilon \cdot a^{\lambda}$ yields exactly Ramanujan's eq. (1). □

This argument follows the standard Mittag-Leffler asymptotic-expansion treatment in [GKMR20]; we cite it both as an independent confirmation of the identity and to make explicit what the §4.4 argument was attempting to construct.

6. Where the §4.4 argument errs

Andrews and Berndt's §4.4 argument proceeds in three steps:

(i) Take the specific cell $\lambda = \frac{1}{2}$, $\varepsilon = 1$.

(ii) Apply Watson's Lemma to the RHS to obtain the asymptotic series [AB13, p. 117, eq. (4.4.1)]:

$$e^a - \frac{1}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n \Gamma(n+1/2)}{a^{n+1/2}}.$$

(iii) Apply Berndt's *Ramanujan's Notebooks, Part I* Chapter 3 Entry 10 [Ber85] to the LHS (with $\varphi(x) = \Gamma(x+1)/\Gamma(x+3/2)$) combined with the gamma-ratio formula [AB13, p. 116, eq. (4.4.2)], yielding [AB13, p. 117, eq. (4.4.3)]:

$$e^a \cdot \left(1 - \frac{3}{8a} + O\left(\frac{1}{a^2}\right)\right).$$

The disagreement at $O(1/a)$ between $e^a - 1/\sqrt{\pi a} + O(1/a^{3/2})$ (the RHS asymptotic from (i)–(ii)) and $e^a - 3e^a/(8a) + O(e^a/a^2)$ (the LHS asymptotic from (iii)) is then taken as proof that the identity fails.

But by section 4, both sides equal $e^a \cdot \operatorname{erf}(\sqrt{a})$ exactly. The asymptotic expansion of $e^a \cdot \operatorname{erf}(\sqrt{a})$ as $a \to \infty$ is

$$e^a \cdot \operatorname{erf}(\sqrt{a}) = e^a \cdot (1 - \operatorname{erfc}(\sqrt{a})) = e^a - \frac{1}{\sqrt{\pi a}} \left(1 - \frac{1}{2a} + \frac{3}{4a^2} - \cdots\right), \tag{9}$$

which *matches* the RHS asymptotic but *contradicts* the LHS asymptotic (4.4.3). The deviation from e^a is exponentially small in a (since $\operatorname{erfc}(\sqrt{a}) \sim e^{-a}/\sqrt{\pi a}$), not polynomial; the leading polynomial coefficient $-3/(8a)$ in (4.4.3) is therefore spurious.

A direct numerical audit at three values of a confirms (4.4.3) is wrong:

a	$e^a \operatorname{erf}(\sqrt{a})$ (true value)	AB (4.4.3): $e^a(1 - 3/(8a))$	rel. err
10	22,026.3	21,200.5	3.75×10^{-2}
100	2.69×10^{43}	2.68×10^{43}	3.75×10^{-3}
1000	1.97×10^{434}	1.97×10^{434}	3.75×10^{-4}

The relative error of $e^a(1-3/(8a))$ decays as $3/(8a)$, exactly the magnitude of the spurious polynomial correction in (4.4.3).

We identify the §4.4 error at three levels:

(i) *Arithmetic.* Applying [AB13, p. 116, eq. (4.4.2)], which is [AS64, 6.147], with $a = 1$, $b = 3/2$ yields the leading correction

$$\frac{(1-3/2)(1+3/2-1)}{2x} = \frac{(-1/2)(1/2)}{2x} = -\frac{1}{8x},$$

not $-3/(8x)$. The coefficient $-3/8$ in (4.4.3) appears to be an arithmetic miscalculation of the application of (4.4.2) with these specific values of the gamma-function parameters.

(ii) *Conceptual.* Berndt's Notebooks Chapter 3 Entry 10 provides the leading-order asymptotic relation between $\sum a^{n+1/2}/\Gamma(n+3/2)$ and a gamma-ratio combination, but the $O(\cdot)$ correction contains hidden polynomial-in-$1/a$ terms that *exactly cancel* against the Stirling expansion of $\Gamma(a+1)/\Gamma(a+3/2)$ when combined to compute the asymptotic of the LHS. The §4.4

argument treats Entry 10 as if its remainder were $O(e^a/a^2)$ rather than $O(e^a/a)$—the leading-cancellation terms are silently dropped, and the spurious $-3/(8a)$ that survives is an artifact of that dropping.

(iii) *Trans-asymptotic mismatch.* Even setting (i) and (ii) aside, comparing a polynomial-truncated expansion (the would-be LHS asymptotic to $O(1/a^2)$) against a divergent Watson-Lemma asymptotic series (the RHS expansion (4.4.1), which is *divergent* in a but resums via Borel–Laplace to give exactly the true value of erfc) is not a valid identity check. The difference between the two sides of eq. (1) is exponentially small in a, which neither asymptotic series captures at any finite truncation.

7. Conclusion

Ramanujan's Entry 4.2.3 (Lost Notebook p. 199) is correct as stated. The §4.4 "Discussion of Entry 4.2.3" in Andrews and Berndt's *Lost Notebook, Part IV* [AB13], and the earlier *Proc. Amer. Math. Soc.* note [Ber04], require correction at the three points identified in section 6.

We propose this short note as a corrigendum. Andrews and Berndt deserve immense credit for their five-volume reconstruction of Ramanujan's *Lost Notebook*—twelve unchallenged years of one specific argument in a single chapter of one volume in no way diminishes that achievement.

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Independent researcher
Email address: `jouni@helminen.co`